EFU: Enforcing Federated Unlearning via Functional Encryption

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01 Context & approach

The challenge

A client's unlearning update matters only if it enters aggregation. *EFU* binds encrypted learning and unlearning updates to an agreed aggregate, while a local objective balances forgetting with retained utility and limited parameter drift.

Our contribution

Bind encrypted updates to an authorized aggregation function.
Combine forget loss, adversarial-example training, and inverted MAS regularization.
Compare sample-, class-, and task-level forgetting with full retraining.

1 Learn or unlearn locally

Train normally or unlearn with the three-term loss; task forgetting also optimizes retained-task loss. Generate adversarial examples once offline.

2 Cluster and encrypt

Compress weights into centroids. Send encrypted centroids, the visible position mapping, and a partial functional key for the round.

3 Decrypt the agreed aggregate

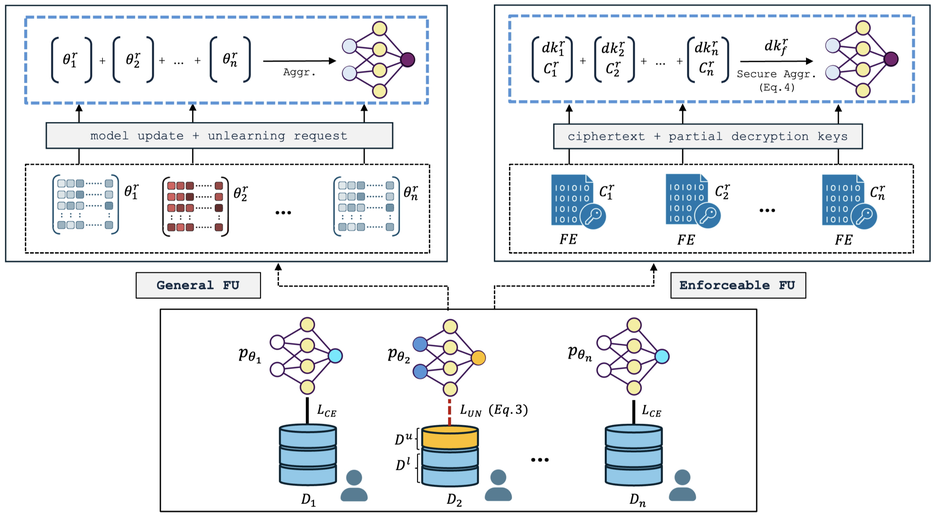
Expand encrypted centroids, combine key shares, and recover the authorized weighted aggregate for the next round.

Scope

Security depends on trusted clients, valid key shares, and the stated indistinguishability assumptions. Low forget accuracy is behavioral evidence. Authorized aggregation does not prove complete erasure or compel model deployment.

02 Method

Bind learning and unlearning updates to an authorized encrypted aggregate.



Core formulation

$$\mathcal{L}_{\text{UN}} = -\mathcal{L}_{\text{CE}}(D^u) + \mathcal{L}_{\text{CE}}(D^{u,\text{adv}}) + \sum_j \tilde{\Omega}_j(\theta_j - \tilde{\theta}_j)^2$$

D^u: forget data; $D^{u,\text{adv}}$: adversarial variants; $\tilde{\theta}$: starting weights; $\tilde{\Omega}$: inverted MAS importance. Forget while limiting unrelated parameter drift.

$$\theta_j^{r+1} = \text{Dec}(\{\theta_{i,\text{enc}}^r[j]\}_i, \text{dk}_f)$$

Encrypted client updates and the function key dk_f yield the agreed aggregate. The binding concerns decryption, not subsequent deployment.

Paper **Code**

03 Results

95.37%
Retained gender accuracy
LFW · task forgetting

5.12%
Forgotten identity accuracy
Same *EFU* model

Class forgetting with retained utility

Method	Retain accuracy (%)	Forget accuracy (%)
Full retrain	94.82	13.72
PGD	91.82	14.31
FedOSD	93.06	12.77
EFU + FedOSD (Ours)	**92.73**	**12.54**
EFU (Ours)	**92.43**	**13.06**

CIFAR-10 · ResNet-18 · forget 10% of classes · 10 clients (1 unlearning) · 20% participation · Dir(0.1). 100 rounds; 10 unlearning rounds from round 50. Means of 3 runs; higher retain / lower forget is better.

Forget identity, retain gender

Method	Gender accuracy (%)	Identity accuracy (%)
Full retrain	96.86	4.71
PGD	94.81	10.32
FedOSD	95.02	5.01
EFU (Ours)	**95.37**	**5.12**

LFW · ViT-B/32 · 30 clients, all unlearning identity · full participation. Ten unlearning rounds from round 50; means of 3 runs.

Takeaway

EFU retains **95.37% gender accuracy** while identity accuracy falls to **5.12%** in LFW task forgetting. Its **authorized encrypted aggregation** incorporates the agreed updates, while complete erasure and subsequent model deployment remain separate requirements.